Atoms and elements

1 Rearrange the letters to make the name we give to the 'basic bits' that all materials are made from.

LEN MEETS _____

2 What is the name of the chart in which all elements and symbols are shown?

P_____ T_____

3 Rearrange these words to give the name of the element whose symbol is shown.

Na	DISUOM	_____
K	MOSSIPAUT	_____
Ag	LIVERS	_____
Pb	DEAL	_____
C	ARCNOB	_____
Si	LISINOC	_____

4 Underline 'true' or 'false' for each statement.

- An element contains different types of atom. **true/false**
- A compound contains atoms of different elements combined together. **true/false**
- An oxide is formed when a metal is burned in oxygen. **true/false**
- Gases are at the far left-hand side of the Periodic Table. **true/false**
- There are about 100 different elements. **true/false**
- More of the elements are non-metals than are metals. **true/false**

5 Complete these sentences using words from the list below.

arranged	**atoms**	**elements**	**materials**	**metals**	**separated**

Wood, sugar and cotton are three _____.

They are made up of the same three _____, carbon, hydrogen and oxygen.

They are different because the _____ of the three elements are _____ in different ways.

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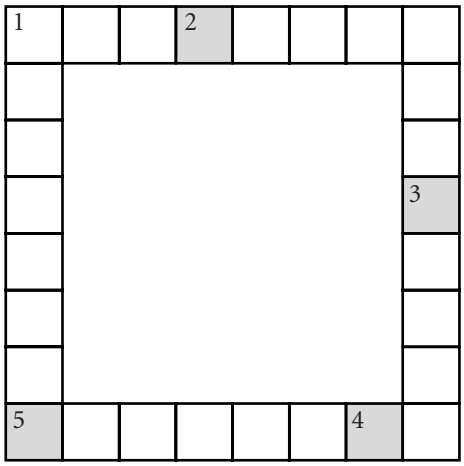
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Atoms and elements continued

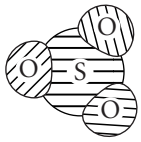
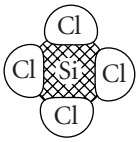
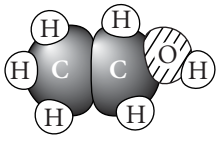
6 Fill the square with the names of five elements. Start at 1 and work clockwise. The shaded squares are at the ends and the beginnings of the names.

Use these clues.

1 A gas that goes in lighting tubes – an element at the top right of the Periodic Table (4 letters)
2 A gas in the air – an element next to oxygen in the Periodic Table (8 letters)
3 A magnetic metal (6 letters)
4 A metal similar in properties to sodium (7 letters)
5 A liquid metal (7 letters)



7 Complete the table.

Name of compound	Diagram of molecule	Number of atoms in molecule
carbon dioxide		3
water		
sulphur trioxide		
ammonia		4
silicon tetrachloride		
ethanol		

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Atoms and elements continued

8 Complete the table.

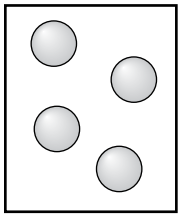
Name of compound	Elements present
copper oxide	copper and oxygen
zinc sulphate	
copper nitrate	
potassium oxide	
potassium chloride	

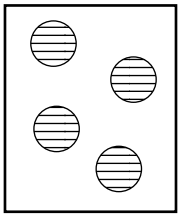
9 Copper sulphide is formed when copper and sulphur react.

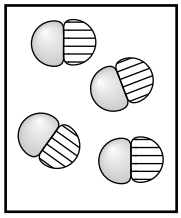
a Write a word equation for this reaction.

_____

b These diagrams show the reaction. What does each represent?







_____ _____ _____

_____ _____ _____

10 Choose from this list of substances to answer the questions.

calcium	**copper oxide**	**magnesium**	**nitrogen**	**iron**
oxygen	**sodium chloride**			

a Which are elements?

_____

b Which are compounds?

_____

c Which are non-metallic elements? _____

d Name one that is an element made of molecules with two atoms. _____

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Compounds and mixtures

1 Rearrange the letters to make the word for a group of atoms in an element or a compound.

O CULL EME _____

2 The formula for glucose is $C_6H_{12}O_6$.

a Is glucose an element, a mixture or a compound? _____

b How many different elements are there in a molecule of glucose? _____

c How many carbon atoms are there in a molecule of glucose? _____

d How many hydrogen atoms are there in a molecule of glucose? _____

e How many oxygen atoms are there in a molecule of glucose? _____

3 Aluminium nitrate has the formula $Al(NO_3)_3$.

a Is aluminium nitrate an element, a mixture or a compound?

_____

b How many elements are there in the formula for aluminium nitrate? _____

c How many aluminium atoms are there in the formula for aluminium nitrate? _____

d How many nitrogen atoms are there in the formula for aluminium nitrate? _____

e How many oxygen atoms are there in the formula for aluminium nitrate? _____

4 Write whether each of the following substances is an element, a mixture or a compound.

oxygen _____ air _____

liquid air _____ pure water _____

mineral water _____ sea water _____

methane gas _____ copper _____

brass _____

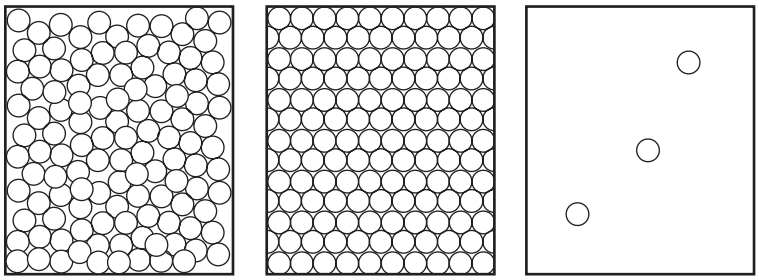
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Compounds and mixtures continued

5 a The three diagrams A, B and C represent the particles in solids, liquids and gases.

Which is which?



A B C

A is a _____

B is a _____

C is a _____

b Complete these sentences by underlining the correct answers, choosing from the bold words.

The particles in B are vibrating. As the temperature rises the particles vibrate faster until eventually the particles break apart and are able to move more freely, as in A.

i This change of state is called **boiling/condensing/freezing/melting**.

ii For water this happens at a temperature of **−5 °C/0 °C/100 °C/105 °C**.

c Complete these sentences by underlining the correct answers, choosing from the bold words.

As the temperature rises, the particles in A move faster and faster until they are able to escape from the other particles and move freely as in C.

i This change of state is called **boiling/condensing/freezing/melting**.

ii For water this happens at a temperature of **−5 °C/0 °C/100 °C/105 °C**.

6 Which of these statements are true and which are false? Cross out the wrong word.

Elements, mixtures and compounds are made up of atoms. true/false

All the atoms in a compound are the same. true/false

The composition of a compound is fixed but that of a mixture is not. true/false

A mixture has a fixed melting point. true/false

In a compound, atoms are joined together. true/false

Continued ⟶

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Compounds and mixtures
continued

7 Read the observations made during experiments.

A Some drops of sodium hydroxide solution were added to a brown solution. A red-brown solid was formed.

B A few crystals of white solid were added to ethanol. The crystals disappeared and a colourless liquid remained.

C Two colourless solutions were mixed and the temperature of the solution fell. Bubbles of a colourless gas were seen.

In which experiments is a chemical reaction likely to have taken place?

_____

8 Amy has two bottles containing the same chemical. In one bottle, labelled X, the chemical is pure and in the other, labelled Y, the chemical is impure.

Tick the statement that is true.

Both melt at the same fixed temperature. ☐

X melts at a lower temperature than Y. ☐

X melts at a fixed temperature and Y over a range of temperature. ☐

Both melt over the same range of temperature. ☐

9 Rearrange the letters to give the names of three methods of separating chemicals.

LOT D INITIALS _____

E TRIFLING _____

VA OPERATION _____

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Rocks and weathering

1 Tick the statements that are true.

A rock is:

a gas	☐	a liquid	☐
a solid	☐	naturally occurring	☐
made by people	☐	living	☐
non-living	☐		

2 Complete the passage using these words. Use each word once only.

cemented	**compounds**	**crystals**	**grains**	**microscope**
minerals	**porous**			

Rocks are made of different chemical _____

called _____. They are found in the rock in tiny bits

called _____. Sometimes these can be seen with the naked

eye, but in some rocks you need a _____ to see them.

In some rocks the tiny bits are _____ together, and in others

they are present as _____ that fit together with no spaces

between them. Rocks that do have spaces between the tiny bits can absorb water

and are said to be _____.

3 Which of these gases dissolves in rainwater to make an acid that attacks rocks? Underline the correct answer.

argon	**carbon dioxide**	**carbon monoxide**	**oxygen**

4 Write down three other things that can cause weathering of rocks.

1 _____

2 _____

3 _____

Continued ⟶

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